

**MATHEMATICS**  
**Introduction to Modern Algebra II**  
**Final.**

1) Let  $f(x)$  and  $g(x)$  be polynomials in  $F[x]$  such that  $(f(x), g(x)) = 1$ . Prove that if  $h(x) = f(x)g(x)$ , then  $F[x]/(h(x)) \cong F[x]/(g(x)) \oplus F[x]/(f(x))$ .

*Proof.* Let  $\phi : F[x] \rightarrow F[x]/(g(x)) \oplus F[x]/(f(x))$  be the homomorphism defined by  $\phi(p(x)) = [p(x) + (f(x)), p(x) + (g(x))]$ . By the First Homomorphism Theorem, it suffices to show that  $\ker(\phi) = (h(x))$  and  $\phi$  is surjective.

The inclusion  $\ker(\phi) \supset (h(x))$  is immediate. Suppose that  $p(x) \in \ker(\phi)$ , then  $p(x) + (f(x)) = 0 + (f(x))$  and  $p(x) + (g(x)) = 0 + (g(x))$  so that  $p(x) \in (f(x))$  and  $p(x) \in (g(x))$ . Thus  $f(x)$  and  $g(x)$  divide  $p(x)$ . But as  $f(x)$  and  $g(x)$  are coprime,  $h(x) = f(x)g(x)$  divides  $p(x)$ , so that  $p(x) \in (h(x))$  as required.

To prove surjectivity, we must show that for any  $[p(x) + (f(x)), q(x) + (g(x))] \in F[x]/(g(x)) \oplus F[x]/(f(x))$ , there exists  $r(x) \in F[x]$  such that  $[r(x) + (f(x)), r(x) + (g(x))] = [p(x) + (f(x)), q(x) + (g(x))]$ . Since  $(f(x), g(x)) = 1$ , we have such that  $1 = f(x)\alpha(x) + g(x)\beta(x)$  for some  $\alpha(x), \beta(x) \in F[x]$ . Thus  $p(x) - q(x) = f(x)\alpha'(x) + g(x)\beta'(x)$  where  $\alpha'(x) = \alpha(x)(p(x) - q(x))$  and  $\beta'(x) = \beta(x)(p(x) - q(x))$ . Let  $r(x) = p(x) - f(x)\alpha'(x) = q(x) + g(x)\beta'(x)$ , then  $\phi(r(x)) = [p(x) + (f(x)), q(x) + (g(x))]$  as required.  $\square$

2) Let  $I \subset R$  be an ideal in a commutative ring with 1. Show that  $I$  is maximal if and only if  $R/I$  is a field.

*Proof.* If  $I$  is maximal and  $r + I \in R/I$  is non-zero, then  $r \notin I$  so that  $(r, I) = R$  and hence  $ar + bi = 1$  for some  $a, b \in R$  and  $i \in I$ . But then  $(a + I)(r + I) = ar + I = (1 - bi) + I = 1 + I$ . Thus all elements in  $R/I^*$  are invertible, and so  $R/I$  is a field.

Conversely, suppose that  $R/I$  is a field and let  $I \subset J \subset R$  be an ideal. We must show that  $J = I$  or  $J = R$ . If  $I \neq J$ , then pick  $j \in J \setminus I$  so that  $j + I \neq 0 + I$ . Since  $R/I$  is a field, there is an element  $k \in R$  such that  $(j + I)(k + I) = 1 + I$ . It follows that  $jk - 1 = i \in I$ . But then  $1 = jk - i \in J + I = J$  and so  $J = R$ .  $\square$

3) Let  $F \subset K$  be an inclusion of fields. Prove that  $k \in K$  is algebraic over  $F$  if and only if  $F(k) \cong F[x]/(f(x))$  where  $f(x) \in F[x]$  is an irreducible polynomial. (Note, that the isomorphism is assumed to fix  $F$ )

*Proof.* Suppose that  $k$  is algebraic over  $F$ , then there is an irreducible polynomial  $f(x) \in F[x]$  such that  $f(k) = 0$ . We may assume that  $f(x)$  has minimal degree amongst all non-zero polynomials such that  $f(a) = 0$ . Let  $\phi : F[x] \rightarrow F(a)$  be the homomorphism defined by  $\phi(g(x)) = g(a)$ . We claim that  $\text{Ker}(\phi) = (f(x))$ . Since  $f(a) = 0$ , we have the inclusion  $\text{Ker}(\phi) \supset (f(x))$ . Suppose on the other hand that  $\phi(g(x)) = 0$  so that  $g(a) = 0$ . Write  $g(x) = f(x)q(x) + r(x)$  where  $\deg(r(x)) < \deg(f(x))$ . We have  $r(a) = g(a) - f(a)q(a) = 0$  and so  $r(x) = 0$ , i.e.  $g(x) \in (f(x))$ . Finally, the image of  $\phi$  is a subfield of  $F(a)$  containing  $F$  and  $a$ , hence  $\phi$  is onto and by the FHT,  $F(a) \cong F[x]/(f(x))$ .

Suppose now that  $F(k) \cong F[x]/(f(x))$ , then  $\dim_F F(k) = n$  where  $n = \deg f(x)$ . But then  $1, k, k^2, \dots, k^n$  are linearly dependent so that  $\sum_{i=0}^n a_i k^i = 0$  for some  $a_i \in F$ . But then  $k$  is a zero of the polynomial  $\sum_{i=0}^n a_i x^i$ .  $\square$

4) Let  $\bar{\mathbb{Q}}$  be the algebraic closure of  $\mathbb{Q}$  in  $\mathbb{C}$ . Show that  $[\bar{\mathbb{Q}} : \mathbb{Q}]$  is infinite.

*Proof.* Let  $a_p$  be a complex root of  $x^p - 2$ . By Eisenstein's Criterion, this polynomial is irreducible and so  $[\mathbb{Q}(a_p) : \mathbb{Q}] = p$ . Since  $\bar{\mathbb{Q}} \supset \mathbb{Q}(a_p) \supset \mathbb{Q}$ , we have  $[\bar{\mathbb{Q}} : \mathbb{Q}] \geq p$  for all  $p > 0$  and hence  $[\bar{\mathbb{Q}} : \mathbb{Q}]$  is infinite.  $\square$

5) Show that  $x^3 + 2x + 2$  is an irreducible polynomial of degree 3 in  $\mathbb{Z}_3[x]$ . Find a generator of the multiplicative group  $(\mathbb{Z}_3[x]/(x^3 + 2x + 2))^*$ . How many generators does this group have?

*Proof.* Since  $x^3 + 2x + 2$  has no roots and has degree 3, it is irreducible. Thus  $\mathbb{Z}_3[x]/(x^3 + 2x + 2)$  is the finite field with  $3^3$  elements and  $(\mathbb{Z}_3[x]/(x^3 + 2x + 2))^*$  is a cyclic group of order 22. There are  $\phi(26) = \phi(2)\phi(13) = 12$  generators.

We have  $(2x + 1)^2 = x^2 + x + 1$ ,  $(2x + 1)^4 = x^2 + 2x = x + 2$ ,  $(2x + 1)^5 = 2x^2 + x + 2$ ,  $(2x + 1)^8 = 2x^2 + 2x + 1$ ,  $(2x + 1)^{13} = 2$  and so the order of  $2x + 1$  is 26.  $\square$

6) Show that  $x^4 + x^3 + x^2 + x + 1$  is irreducible in  $\mathbb{Q}(\sqrt[5]{3})[x]$ .

*Proof.*  $\phi_4(x) = x^4 + x^3 + x^2 + x + 1$  is irreducible in  $\mathbb{Q}[x]$  (as seen in class). Let  $\theta$  be a root of  $\phi_4(x)$ . Then  $[\mathbb{Q}(\theta) : \mathbb{Q}] = 4$ . Since  $x^5 - 3$  is irreducible (Eisenstein's Criterion), we have  $[\mathbb{Q}(\sqrt[5]{3}) : \mathbb{Q}] = 5$ . Since 4 and 5 are coprime,  $[\mathbb{Q}(\theta, \sqrt[5]{3}) : \mathbb{Q}] = 20$ . But then  $20 = [\mathbb{Q}(\theta, \sqrt[5]{3}) : \mathbb{Q}(\sqrt[5]{3})] \cdot [\mathbb{Q}(\sqrt[5]{3}) : \mathbb{Q}]$  so that  $[\mathbb{Q}(\theta, \sqrt[5]{3}) : \mathbb{Q}(\sqrt[5]{3})] = 4$ , and hence  $x^4 + x^3 + x^2 + x + 1$  is irreducible in  $\mathbb{Q}(\sqrt[5]{3})[x]$ .  $\square$

7) Prove that  $[\mathbb{Q}(\sqrt{2} + \sqrt{3}) : \mathbb{Q}] = 4$

*Proof.* Let  $a = \sqrt{2} + \sqrt{3}$ , then  $a^2 = 5 + 2\sqrt{6}$  and  $a^3 = 5a + 4\sqrt{3} + 3\sqrt{2} = 8a + \sqrt{3}$ . But then  $\sqrt{3} \in \mathbb{Q}(\sqrt{2} + \sqrt{3})$  and hence  $\sqrt{2} \in \mathbb{Q}(\sqrt{2} + \sqrt{3})$  so that  $\mathbb{Q}(\sqrt{2} + \sqrt{3}) = \mathbb{Q}(\sqrt{2}, \sqrt{3})$ . It follows that  $[\mathbb{Q}(\sqrt{2} + \sqrt{3}) : \mathbb{Q}] = [\mathbb{Q}(\sqrt{2}, \sqrt{3}) : \mathbb{Q}(\sqrt{3})][\mathbb{Q}(\sqrt{3}) : \mathbb{Q}] \leq 4$ . To show equality, it suffices to show that  $\sqrt{2} \notin \mathbb{Q}(\sqrt{3})$ . Suppose  $\sqrt{2} = a + b\sqrt{3}$  where  $a, b \in \mathbb{Q}$ , then  $2 = a^2 + 3b^2 + 2ab\sqrt{3}$  and so  $\sqrt{3} = \frac{2-a^2-3b^2}{2ab} \in \mathbb{Q}$  which is impossible (if  $ab \neq 0$ ,  $\sqrt{3} \in \mathbb{Q}$  which is impossible; if  $ab = 0$  then either  $b = 0$  so that  $\sqrt{2} = a$  or  $a = 0$  so  $\sqrt{2} = b\sqrt{3}$  and both cases are easily seen to be impossible).  $\square$

8) Show that if  $n \geq 3$ , then  $A_n$  is generated by 3-cycles.

*Proof.* Let  $t = (i, j)$  and  $t' = (i', j')$  be transpositions. We may assume  $i < j$  and  $i' \leq j'$ . We claim that if  $t \neq t'$ , then  $tt'$  is a product of two 3-cycles. If  $i, i', j, j'$  are all distinct, then  $(i, j)(i', j') = (i, j, i')(i', j', j)$ . If  $i = i'$ , then  $(i, j)(i, j') = (i, j', j) = (i, j, j')(i, j, j')$ .

If  $g \in A_n$ , then  $g = t_1 \cdots t_{2k}$  where  $t_i$  are transpositions. So the first claim follows since by what we have shown above, we have  $t_{2i}t_{2i+1} = g_{2i}g_{2i+1}$  where  $g_{2i}$  and  $g_{2i+1}$  are 3-cycles.  $\square$